

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2059

Snapper Creek 138 kV
November 28, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2059. The Interconnection Customer proposes a 144.2 MW battery energy storage facility interconnecting at the Snapper Creek 138 kV substation in the Pennsylvania area. The request was received on April 08, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 564 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 32 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$104,745,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pulpgen-Northampton Generating Company Lp 138.0 kV	St Hedwig-Snapper Creek 138.0 kV	149.2	185	52.5
Pulpgen-Clear Crossing 138.0 kV	Clear Springs 500/345 kV Transformer Bank	134.8	330	13.3
Snapper Crk-St Hedwig 138.0 kV line	Pulpgen Substation-Northampton Generating Company Lp 138.0 kV	133.0	244	84.2
Walden North Generation Station-Los Angeles Hts 138.0 kV	Clear Springs 500 kV/345 kV Auto	132.2	369	13.3
Van Wagner-Los Angeles Hts 138.0 kV	Clear Springs 500 kV/345 kV Transformer	128.2	388	13.3
Landis Delivery No 2-East Arlington 138.0 kV L	Edwardsport Station-Mount Olive 138.0 kV	127.8	418	5.0
Okawville-Landis Delivery No 2 138.0 kV line	Mount Olive-Edwardsport Sta 138.0 kV	127.0	428	5.0
Boscobel Muni-S Lane City 138.0 kV Line	White Swan-S Mtn 138.0 kV	122.7	262	5.1
S Mountain-White Swan 138.0 kV	Equon Substation-Kooskia 138.0 kV	121.7	280	5.6
Morgan Henderson Substation-Bell Substation 138.0 kV	Polk St-Railhead 138.0 kV	120.3	254	10.3
Esquon-Kooskia 138.0 kV line	White Swan-South Mountain 138.0 kV Line	120.0	280	5.1
Polk St-Van Wagner 138.0 kV	W Gallatin-Walnut 345.0 kV L	119.4	445	8.0
Northampton Generating Company Lp Substation-Hogal 138.0 kV	St Hedwig-Snapper Crk 138.0 kV	118.4	215	52.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Scitico-Covelo 138.0 kV line	Pulpen-Northampton Generating Company Lp 138.0 kV	117.1	218	15.8
Walden N Generation Sta Substation-Windstone 138.0 kV L	Clear Springs 500/345kV Autotransformer	116.9	403	13.3
Windstone-Clear Crossing 138.0 kV	Clear Springs 500/345 kV TX	116.9	390	13.3
S Lane City-Equon 138.0 kV	White Swan-South Mtn 138.0 kV	116.9	279	5.1
Rec Laprairie Substation-Sumner 138.0 kV	Van Wagner-Los Angeles Heights 138.0 kV	116.5	281	24.8
Bell-Sanborn 138.0 kV	Polk St-Railhead 138.0 kV	115.9	245	10.3
Saint Hedwig-Steel City 138.0 kV	Northampton Generating Company Lp-Pulpgen 138.0 kV	113.8	264	84.2
Pulpgen-Covelo 138.0 kV	Pulpgen-Northampton Generating Company Lp 138.0 kV	112.9	258	15.8
Van Wagner-Sanborn 138.0 kV	Plk St-Railhead 138.0 kV	112.4	234	10.3
Pol St-Railhead 138.0 kV Line	W Gallatn-Wlnut 345.0 kV	111.6	489	8.0
Geigy-Hollywood 138.0 kV	Van Wagner-Los Angeles Heights 138.0 kV	111.6	220	24.8
Wescosville-East Arlington 138.0 kV line	Landis Delivery No 2-Okawville 138.0 kV	109.5	434	5.7
Dornyville-Wescosville 138.0 kV L	Landis Delivery No 2-Okawville 138.0 kV	108.8	453	5.7
Railhead-Boscobel Muni 138.0 kV	South Mtn-White Swan 138.0 kV	108.1	283	5.1
Dornyvle-Edwardsport Sta 138.0 kV	Landis Delivery No 2-Okawville 138.0 kV	108.0	478	5.7
Edwardsport Sta Substation-Mount Olive 138.0 kV	Okawville Substation-Landis Delivery No 2 138.0 kV	107.3	503	5.7
San German 6403-Hogal 138.0 kV	Saint Hedwig-Snaper Crk 138.0 kV	105.7	229	52.5
Guanica Tc-Rialto 138.0 kV L	Pulpgen-Northampton Generating Company Lp 138.0 kV	105.4	173	24.2
Steel City-Guanica Tc Substation 138.0 kV	Pulpgen-Northampton Generating Company Lp Substation 138.0 kV	100.5	234	24.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pulpgen-Northampton Generating Company Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$2,064,000	100% (\$2,064,000)
Pulpgen-Clear Crossing 138.0 kV	Rebuild line with higher-capacity conductor	\$10,144,000	100% (\$10,144,000)
Snapper Creek-St Hedwig 138.0 kV	Rebuild line with higher-capacity conductor	\$2,045,000	100% (\$2,045,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Walden North Generation Station-Los Angeles Hts 138.0 kV	Rebuild line with higher-capacity conductor	\$5,764,000	100% (\$5,764,000)
Los Angeles Heights-Van Wagner 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,921,000	100% (\$5,921,000)
Landis Delivery No 2-East Arlington 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,712,000	7% (\$407,000)
Okawville-Landis Delivery No 2 138.0 kV line	Rebuild line with higher-capacity conductor	\$5,554,000	7% (\$395,000)
Boscobel Muni-S Lane City 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,316,000	100% (\$5,316,000)
White Swan Substation-South Mtn 138.0 kV L	Rebuild line with higher-capacity conductor	\$7,072,000	100% (\$7,072,000)
Morgan Henderson-Bell 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,536,000	100% (\$3,536,000)
Esquon-Kooskia 138.0 kV L	Rebuild line with higher-capacity conductor	\$3,777,000	100% (\$3,777,000)
Polk St-Van Wagner 138.0 kV	Rebuild line with higher-capacity conductor	\$5,575,000	100% (\$5,575,000)
Hogal-Northampton Generating Company Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$2,601,000	100% (\$2,601,000)
Covelo-Scitico 138.0 kV L	Rebuild line with higher-capacity conductor	\$917,000	100% (\$917,000)
Walden N Generation Sta Substation-Windstone 138.0 kV L	Rebuild line with higher-capacity conductor	\$5,010,000	100% (\$5,010,000)
Clear Crossing-Windstone Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,018,000	100% (\$3,018,000)
S Lane City-Equon 138.0 kV	Rebuild line with higher-capacity conductor	\$9,520,000	100% (\$9,520,000)
Sumner-Rec Laprairie 138.0 kV	Rebuild line with higher-capacity conductor	\$4,495,000	68% (\$3,065,000)
Bell-Sanborn 138.0 kV	Rebuild line with higher-capacity conductor	\$2,684,000	100% (\$2,684,000)
Saint Hedwig-Steel City 138.0 kV line	Rebuild line with higher-capacity conductor	\$590,000	100% (\$590,000)
Pulpgen-Covelo 138.0 kV	Rebuild line with higher-capacity conductor	\$6,770,000	100% (\$6,770,000)
Van Wagner-Sanborn 138.0 kV	Rebuild line with higher-capacity conductor	\$6,759,000	100% (\$6,759,000)
Polk St-Railhead 138.0 kV	Rebuild line with higher-capacity conductor	\$4,677,000	100% (\$4,677,000)
Geigy-Hollywood 138.0 kV	Rebuild line with higher-capacity conductor	\$736,000	68% (\$502,000)
Wescosville-East Arlington 138.0 kV line	Reconductor line	\$502,000	15% (\$75,000)
Dornyvile-Wescosville 138.0 kV	Reconductor line	\$968,000	15% (\$145,000)
Railhead-Boscobel Muni 138.0 kV	Reconductor line	\$2,646,000	100% (\$2,646,000)
Dornyvile-Edwardsport Sta 138.0 kV	Reconductor line	\$3,467,000	15% (\$520,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MountOlive-Edwardsport Sta 138.0 kV	Reconductor line	\$616,000	15% (\$92,000)
San German 6403-Hogal Substation 138.0 kV	Reconductor line	\$1,439,000	100% (\$1,439,000)
Guanica Tc-Rialto 138.0 kV L	Reconductor line	\$800,000	81% (\$651,000)
Steel City-Guanica Tc Substation 138.0 kV	Reconductor line	\$1,289,000	81% (\$1,048,000)
	Total allocated to project		\$104,745,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.